IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25_26J_009

1. Topic

A Lightweight Trust Management Framework in the Networking Layer of IoT Devices

2. Research group the project belongs to

CI - Computing Infrastructure

3. Specialization of the project belongs to

Computer Systems and Network Engineering (CSNE)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references – references not included in word count.

The Internet of Things (IoT) has transformed modern infrastructure monitoring, particularly in critical applications such as disaster management and smart city deployments. However, the multi-hop nature of IoT networks, where data traverses through multiple intermediate nodes before reaching its destination, creates significant security vulnerabilities [1]. Current RPL (IPv6 Routing Protocol for Low-Power and Lossy Networks) implementations are susceptible to various attacks including rank attacks, sinkhole attacks, selective forwarding, and insider threats, which can compromise data integrity and network availability [2][3].

Existing trust management solutions for RPL suffer from several critical limitations [4]. First, they impose high computational and energy overhead, making them unsuitable for resource-constrained IoT devices [5]. Second, current approaches lack dynamic adaptation capabilities, relying on static trust models that cannot respond to changing network conditions or evolving attack patterns [6]. Third, most solutions address individual attack types rather than providing comprehensive security frameworks [7]. Finally, existing trust mechanisms often require centralized control, creating single points of failure and scalability issues [8][9].

The challenge is particularly acute in disaster scenarios where IoT sensor networks must operate reliably under adverse conditions [10]. For instance, earthquake monitoring systems require trustworthy data from distributed sensors to enable timely emergency response [11]. However, the inability to distinguish between legitimate sensor malfunctions and malicious attacks can lead to false alarms or missed critical events, potentially resulting in catastrophic consequences.

This study responds to the problem of creating a lightweight, distributed trust management framework applicable at the network layer of IoT devices. The IoT nodes need to provide real-time evaluation and detection of trust breaches and attacks as well as balancing security requirements with limited resources. This study aims to develop basic trust mechanisms for securing IoT data transmission at the network level through RPL so that they do not exceed the resource-constrained device limits.

References:

[1] D. Airehrour, J. A. Gutierrez, and S. K. Ray, "SecTrust-RPL: A secure trust-aware RPL routing protocol for Internet of Things," Future Generation Computer Systems, vol. 93, pp. 860-876, Apr. 2019.

[2] M. Zaminkar and R. Fotohi, "SoS-RPL: Securing Internet of Things against sinkhole attack using RPL protocol-based node rating and ranking mechanism," arXiv preprint arXiv:2005.09140, May 2020.

[3] T. Winter et al., "RPL: IPv6 routing protocol for low-power and lossy networks," RFC 6550, Internet Engineering Task Force, Mar. 2012.

[4] C. Cervantes, D. Poplade, M. Nogueira, and A. Santos, "Detection of sinkhole attacks for supporting secure routing on 6LoWPAN for Internet of Things," in IFIP International Conference on Autonomous Infrastructure, Management and Security, 2015, pp. 91-103.

[5] A. Verma and V. Ranga, "Security of RPL based 6LoWPAN networks in the Internet of Things: A review," IEEE Sensors Journal, vol. 20, no. 11, pp. 5666-5690, Jun. 2020.

[6] P. O. Kamgueu, E. Nataf, and T. Djotio Ndie, "A survey on RPL enhancements: A focus on topology, security and mobility," Computer Communications, vol. 120, pp. 10-21, May 2018.

[7] N. Djedjig, D. Tandjaoui, F. Medjek, and I. Romdhani, "New trust metric for the RPL routing protocol," in Proc. 2017 8th Int. Conf. Information and Communication Systems (ICICS), Irbid, Jordan, 2017, pp. 328-335.

[8] S. Y. Hashemi and F. Shams Aliee, "Fuzzy, dynamic and trust based routing protocol for IoT," Journal of Network and Systems Management, vol. 28, no. 4, pp. 1533-1562, Oct. 2020.

[9] K. A. Awan, I. U. Din, A. Almogren, M. Guizani, and S. Khan, "StabTrust-a stable and centralized trust-based clustering mechanism for IoT enabled vehicular ad-hoc networks," IEEE Access, vol. 8, pp. 21159-21177, 2020.

[10] F. A. Alaba, M. Othman, I. A. T. Hashem, and F. Alotaibi, "Internet of Things security: A survey," Journal of Network and Computer Applications, vol. 88, pp. 10-28, Jun. 2017.

[11] S. Chen, H. Xu, D. Liu, B. Hu, and H. Wang, "A vision of IoT: Applications, challenges, and opportunities with China perspective," IEEE Internet of Things Journal, vol. 1, no. 4, pp. 349-359, Aug. 2014.

6. Brief description of the nature of the solution including a conceptual diagram

The proposed solution is a **Lightweight Trust Management Framework for RPL-based IoT Networks** that ensures intelligent behavior monitoring alongside adaptive response mechanisms. As illustrated in the conceptual diagram, the framework functions within an IoT cluster set-up where sensor nodes are always vigilant of network activity and monitor behaviors while a root node acts as the network head and coordinator.

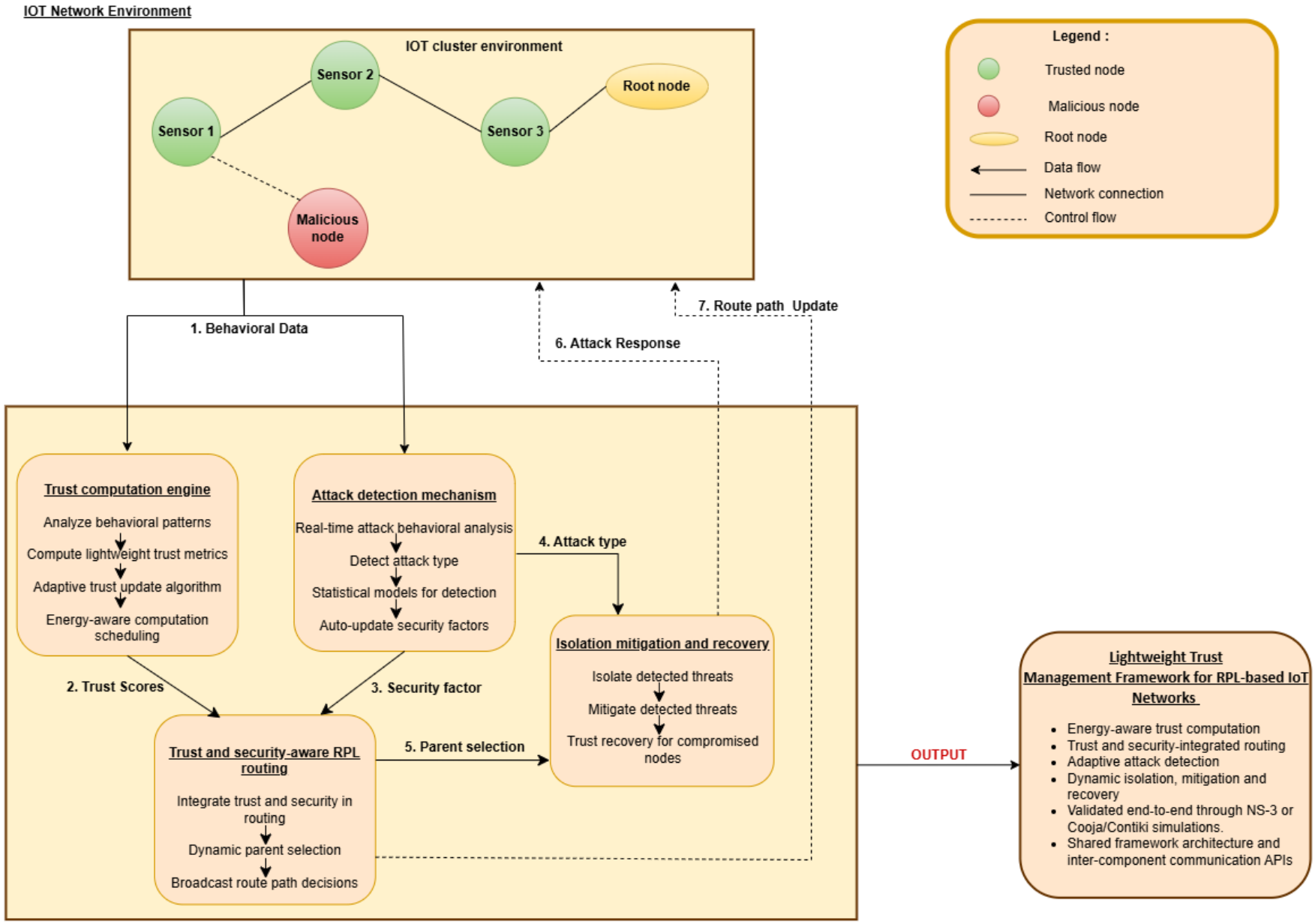
The **Trust Computation Engine** processes behavioral data from sensor nodes to compute lightweight trust metrics using energy-aware algorithms and adaptive trust update mechanisms, which collectively form trust scores. These trust scores are incorporated into a Trust and Security RPL Routing module capable of dynamic parent selection and broadcasting secure route path decisions all while ensuring data transmission only occurs among verified trusted nodes.

Real-time behavior analysis combined with statistical methods enables instant updates to security levels upon detection of specified threats through the **Attack Detection Mechanism**, which offers multi-faceted attack detection capabilities.

Upon detection, the system initiates **isolation, mitigation, recovery** processes that implement targeted threat removal strategies while restoring compromised nodes' status within the trusted framework and mitigating them thereby isolating detected threats.

The operational sequence begins with behavioral data collection followed by trust score calculation, security factor analysis, identification of the attack type, parent selection optimization, attack response deployment and finally route path updates. The result is an autonomous self-protecting ecosystem that provides and maintains network security for resource-constrained IoT devices.

As proven by NS-3 and Cooja/Contiki simulations, the solution offers energy-aware trust computation, integrated security routing, adaptive attack detection as well as dynamic recovery mechanisms. The provided framework gives shared architecture along with inter-component communication APIs which makes it possible to implement critical applications such as disaster management.


IOT Network Environment
IOT cluster environment
Sensor 1
Sensor 2
Sensor 3
Root node
Malicious node
Legend :
Trusted node
Malicious node
Root node
Data flow
Network connection
Control flow
1. Behavioral Data
6. Attack Response
7. Route path Update
Trust computation engine
Analyze behavioral patterns
Compute lightweight trust metrics
Adaptive trust update algorithm
Energy-aware computation scheduling
Attack detection mechanism
Real-time attack behavioral analysis
Detect attack type
Statistical models for detection
Auto-update security factors
4. Attack type
Isolation mitigation and recovery
Isolate detected threats
Mitigate detected threats
Trust recovery for compromised nodes
2. Trust Scores
3. Security factor
Trust and security-aware RPL routing
Integrate trust and security in routing
Dynamic parent selection
Broadcast route path decisions
5. Parent selection
OUTPUT
Lightweight Trust Management Framework for RPL-based IoT Networks
Energy-aware trust computation
Trust and security-integrated routing
Adaptive attack detection
Dynamic isolation, mitigation and recovery
Validated end-to-end through NS-3 or Cooja/Contiki simulations.
Shared framework architecture and inter-component communication APIs

7. Brief description of specialized domain expertise, knowledge, and data requirements

Specialized Domain Expertise Required:

Network Protocol Design: Comprehensive knowledge in RPL protocol specifications such as DODAG construction, objective functions, and control messages formatting. Familiarity with addressing schemes in IPv6, 6LoWPAN header compression, and other low power wireless communication protocols.

IoT Security: Knowledge of tailored attack vectors for IoT ecosystems, trust management frameworks, and lightweight cryptographic algorithms. Consideration of resource constraints like processing capabilities, memory restrictions, and energy consumption within IoT devices.

Trust Management Systems: Understanding of trust computation models encompassing trust propagation mechanisms and dynamic adaptation computation algorithms. Known behavioral analysis for anomaly detection through various statistical approaches.

Simulation and Modeling: Expertise in network simulation (NS-3, Cooja/Contiki-NG) and implementation of IoT protocols. Experience working with performance evaluation frameworks alongside statistical methods.

Knowledge Requirements:
Programming Languages: Active knowledge includes C/C++ for protocol implementation; Python for handling data sets by analyzing or via machine learning algorithms. Concepts embedded systems programming along with real-time operating systems concerning IoT devices is essential.

Mathematical foundations: Statistics applies to trust computation while graph theory is dedicated toward analyzing network topologies alongside optimization theory focusing on designing routing algorithms.

Data Requirements:

Primary Data:

- Simulation datasets generated from NS-3 or Cooja environments including network topology data, traffic patterns, and attack scenarios.
- IoT-23 Dataset: https://www.stratosphereips.org/datasets-iot23 (IoT network traffic with malicious and benign behavior)
- UNSW-NB15 Dataset: https://research.unsw.edu.au/projects/unsw-nb15-dataset (Network intrusion detection dataset)

Secondary Data:

- Intel Lab Data: http://db.csail.mit.edu/labdata/labdata.html (Real sensor network deployment data)
- RPL Implementation benchmarks from Contiki-NG: https://github.com/contiki-ng/contiki-ng

Validation Data:

- CRAWDAD IEEE 802.15.4 datasets: https://crawdad.org/
- Smart City datasets from Santander: http://smartsantander.eu/
- Earthquake sensor data from USGS: https://earthquake.usgs.gov/data/

All data will be gathered through controlled simulation experiments as well as publicly available datasets, safeguarding ethical considerations of data use. This approach enhances reproducibility and credibility of the research outcomes.

8. Objectives and Novelty

Main Objective: Develop a Lightweight Trust Management Framework for RPL-based IoT Networks			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Kaluarachchi D.N. IT22144362	Design energy efficient trust computation engine.	- Develop lightweight trust metrics based on behavioral patterns - Create adaptive trust update algorithms - Implement energy-aware computation scheduling. - Design a framework architecture and APIs for inter-component communication and deployment of a simulation environment.	Trust evaluation based on energy-aware systems which modify the frequency of computations according to the battery status of the node and prevailing network conditions.
Wijekoon H.M.R.K. IT22065162	Implement trust and security aware RPL routing mechanism.	- Modify RPL objective function to integrate trustworthiness and security factor - Implement a dynamic parent selection algorithm - Broadcast route path decisions. - Design a framework architecture and APIs for inter-component communication and deployment of a simulation environment.	Balanced objective function that maintains RPL performance, integrating trust scores and security factors for routing decisions.

Wijetunga S.S. IT22342126	Develop secure attack detection mechanism.	- Identifying real-time attack behavioral analysis and detect attack type - Create statistical models for attack detection - Mechanism to auto update the security factor and repair trust relationship. - Design a framework architecture and APIs for inter-component communication and deployment of a simulation environment.	Attack-specific behavioral analysis and security factor with adaptive detection thresholds that minimize false positives in dynamic IoT environments.
Bandara R.M.T. IT18065640	Design isolation, mitigation and recovery mechanisms.	- Design response mechanisms for isolate detected threats - Mechanisms for mitigating detected threats - Develop trust recovery mechanisms for failed/compromised nodes. - Design a framework architecture and APIs for inter-component communication and deployment of a simulation environment.	Unified trust-aware response mechanism which isolates malicious nodes, mitigation in real time and recovery using lightweight trust secure relationship.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Kaluarachchi D.N. IT22144362	Concentrates on network-layer security through trust computation algorithms designed for RPL protocol. Since it's a Lightweight Trust Management Framework, it involves deep understanding of IoT network architectures, energy-efficient algorithm design, and resource constrained computing. The component directly addresses the problems of security in IoT systems and consequently needs knowledge in distributed systems, protocol design, and security mechanisms. The energy-aware trust computation integrates well with principles of optimization and resource management within networking specialization.
Wijekoon H.M.R.K. IT22065162	Focuses on alterations made to a routing protocol and enhances the performance of the network. Entails in-depth knowledge of RPL protocol internals, objective function design, as well as topology control for networks. This component requires an expert understanding of routing algorithms as well as performance analysis concerning networks and engineering protocols. Trust-aware routing links network security issues substantially with performance optimization which are fundamental areas within networking specialization. Incorporating trust metrics into decisions demonstrates advanced skills in networking protocol design through refined decision-making processes.
Wijetunga S.S. IT22342126	Concentrates on network security concerning attack detection and threat mitigation at the network level. Requires knowledge of network security analyses, intrusion detection systems, and behavioral analysis methods. This component includes the study of different network attacks, modeling security threats, and real-time detection systems. The statistical and behavioral analysis approaches together with security surveillance constitute a specialization in network security requiring expertise in monitoring, threat intelligence, and incident response operations.
Bandara R.M.T. IT18065640	Focuses on trust information management and network resilience mechanisms. Involve expertise in distributed protocol design as well as in fault tolerance and network recovery systems. The component requires understanding of trust propagation algorithms, network reliability, and distributed consensus mechanisms. Trust recovery and propagation mechanisms align with network resilience and systems aspects of networking specialization that demonstrate advanced knowledge of network fault tolerance principles and system design.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Shashika	Lokuliyana	
Co-Supervisor	Ms.	Pipuni	Wijesiri	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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Topic Assessment Form

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.